CS341 · COMPUTER GRAPHICS

Calculations Practice

Practice on every calculation type

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01 Questions

Q1 ASPECT RATIO

a) A screen is 12 inches wide and 9 inches high with a resolution of 1024 × 768. Find the aspect ratio.

b) A screen is 10 inches wide and 8 inches high with a resolution of 800 × 600. Find the aspect ratio.

Q2 RESOLUTION

A monitor measures 32 cm × 24 cm and shows 1280 × 960 pixels. Find the resolution in pixels per cm in x and y, and the aspect ratio.

Q3 FRAME BUFFER

How much frame-buffer memory (in bytes) does a 1024 × 768 screen need with:
a) 1 bit per pixel b) 8 bits per pixel c) 24 bits per pixel

Q4 VECTORS

a) P = (2, 5), Q = (7, 1). Find the vector from P to Q and its length.

b) a = (1, 2, 2). Find |a| and the unit vector â.

c) b = (2, −1, 0). Find a · b. Are a and b perpendicular?

d) Find a × b.

e) Find the angle between u = (1, 0) and w = (−1, 1).

f) Find the counterclockwise perpendicular of c = (3, −2).

Q5 POINTS & LINES

a) Which are valid points? $0.5P + 0.5R$, $2P - R$, $P + R$, $0.2P + 0.3R$

b) $A = (1, 2)$, $B = (9, 6)$. Find the point at $t = 0.75$ on $P = A + t(B - A)$.

c) For $C = (2, 1)$ and $B = (5, 5)$, write the parametric form and the point-normal form of the line.

Q6 COLOR MODELS

a) Convert RGB (0.2, 0.6, 1) to CMY.

b) Convert CMY (0, 1, 1) to RGB. What color is it?

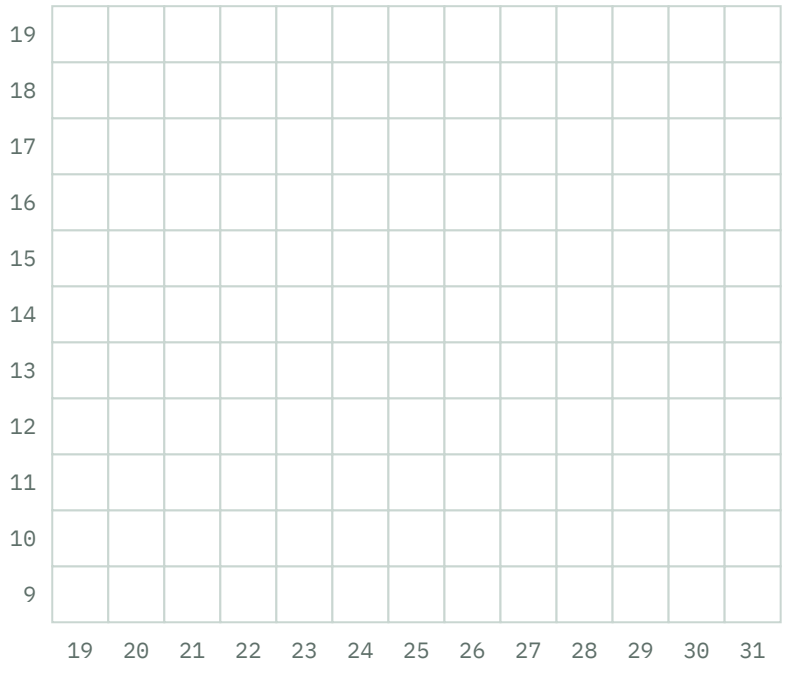
c) Find the YIQ luminance Y of green (0, 1, 0) and magenta (1, 0, 1).

d) Write the HSV values of green, cyan and blue, and the HLS value of red.

Q7 DDA

Using DDA, find the pixels from (20, 10) to (30, 18).

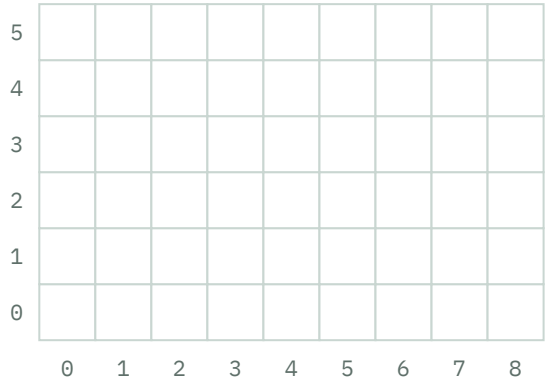
k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				



Q8 DDA

Using DDA, find the pixels from (1, 1) to (7, 4).

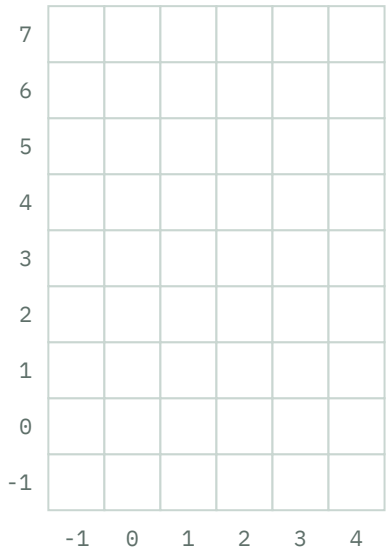
k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				
6				



Q9 DDA · |m| > 1

Using DDA, find the pixels from (0, 0) to (3, 6).

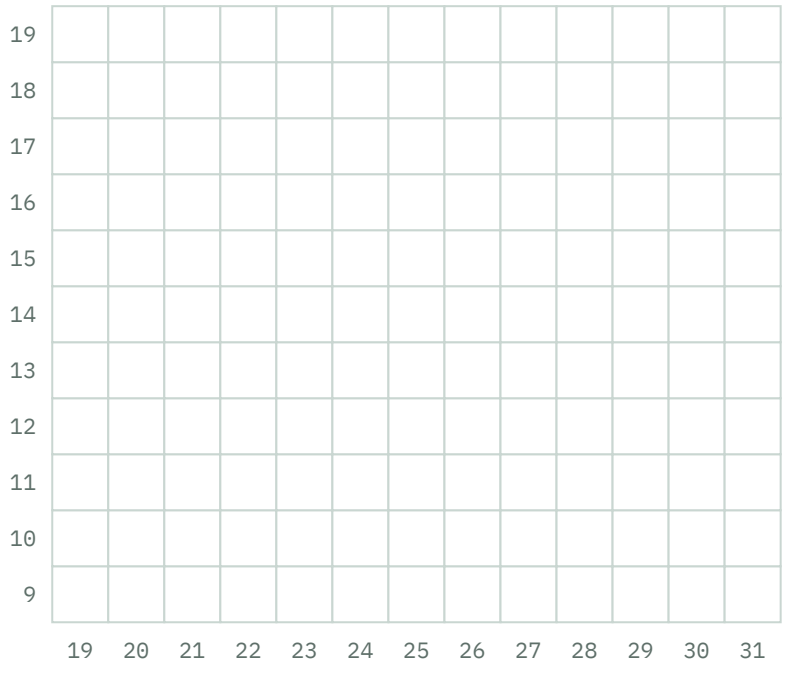
k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				
6				



Q10 BRESENHAM

Using Bresenham's algorithm, find the pixels from (20, 10) to (30, 18).

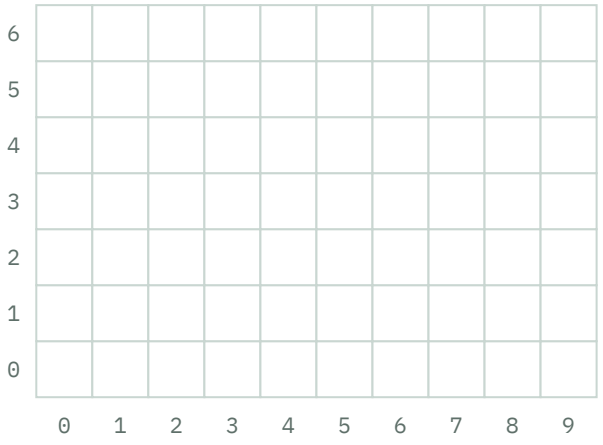
k	p_k	Decision	Next pixel	p_{k+1}
0				
1				
2				
3				
4				
5				
6				
7				
8				
9				



Q11 BRESENHAM

Using Bresenham's algorithm, find the pixels from (1, 1) to (8, 5).

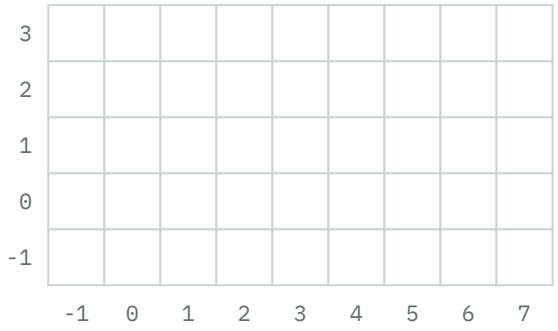
k	p_k	Decision	Next pixel	p_{k+1}
0				
1				
2				
3				
4				
5				
6				



Q12 BRESENHAM

Using Bresenham's algorithm, find the pixels from (0, 0) to (6, 2).

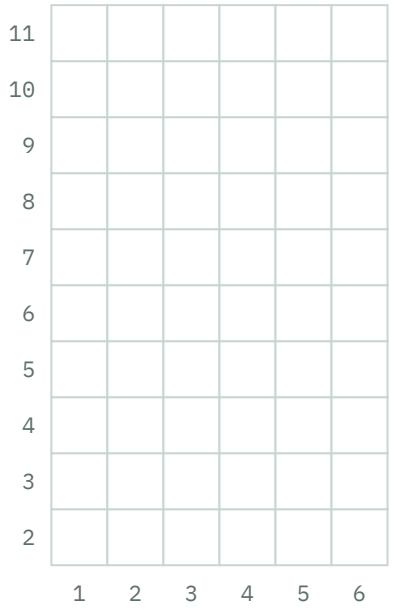
k	p_k	Decision	Next pixel	p_{k+1}
0				
1				
2				
3				
4				
5				



Q13 BRESENHAM · m > 1

Using Bresenham's algorithm, find the pixels from (2, 3) to (5, 10).

k	p_k	Decision	Next pixel	p_{k+1}
0				
1				
2				
3				
4				
5				
6				



Q14 CIRCLE

Using Bresenham's circle algorithm, find the pixels of the first (top) octant of a circle centered at the origin with r = 8.

Q15 CIRCLE

Using Bresenham's circle algorithm, find the pixels of the first (top) octant for r = 10, then write the 8 symmetric points of (3, 10).